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PAPER

Design and parametrization of TPMS lattice using computational and experimental approach

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Abstract

Triply periodic minimal surface (TPMS) based lattices are extensively explored as a scaffold design for bone regeneration. TPMS maintains zero mean curvature at each point and offers a large surface area comparable to a trabecular bone. The best four TPMS minimal surfaces (IWP. Neovius, primitive, and F-RD) were selected, designed, and fabricated using acrylonitrile butadiene styrene (ABS) resin through the stereolithography (SLA) technique. The results indicate that small changes in unit cell dimensions do not significantly alter the structure topology, which ensures stress distribution within the lattice remains relatively uniform across different unit cell sizes when the porosity level is constant. The optimal unit cell size (2 to 5 mm) and porosity (70 to 80%) significantly affect the compressive strength and surface area to volume (SA/V) ratio due to a unique arrangement of the internal architecture of each TPMS unit cell. The lattice structure (formed by stacking unit cell) of unit cell size 2.11 mm with 70% porosity exhibited a maximum compressive strength of 39.8 MPa in IWP, followed by Neovius, primitive, and F-RD-based lattice structures. Moreover, the lattice showed more stability under compression force, minimized stress concentration compared to a unit cell, and exhibited distinct deformation patterns at different strain levels during compression.

1. Introduction

Bones are vital in providing stability and framework to the human body. This function is facilitated by the presence of two distinct types of bone tissue: compact bone and spongy bone. These bones have non-uniform porous cross-sections with varying porosity from 5% to 95%, depending upon the types and function of the bone. The bones can be categorized based on their shapes, such as flat bones, long bones (femur, tibia), short bones, irregular bones, and sesamoid bones, as shown in appendix, figure A1 [1]. Compact bone generally serves as the robust and dense tissue that imparts rigidity and unyielding nature to the bones. It supports the femur, tibia, and the rest of the body's long bones and accounts for approximately 80% of the weight of the skeleton system. On the other hand, spongy bone, also called trabecular bone, consists of interconnected trabeculae, slender and rod-like structures made of connective tissue. It can be found in the distal end of the radius, the proximal humerus, or the proximal femur. The structural and mechanical properties of bone tissues are provided in appendix, table A1 [2]. The cortical bone exhibited a more prominent Young's modulus value (7–30 GPa), providing adequate mechanical strength (100–230 MPa) to support loads. Conversely, cancellous bone has a lower Young's modulus value (0.05–0.5 GPa) than cortical bone but has a high specific surface area (20 mm^2 mm^{-3}) and can dampen sudden stress. The human body weight is supported on the leg by the femur bone, which makes us stand, walk, and run. The femur bone has different levels of porosity in its various parts. In appendix, figure A2 shows a femur bone with different sections along the length and the level of porosity available in other parts for two male and female candidates [3]. The most frequent problem associated with the femur is a fracture, which is more likely in the elderly due to loss of bone density but also prevails in other age groups due to its load-bearing virtue. Fractures can be classified into different types based on location, pattern, and severity. In appendix, figure A3 shows some commonly observed fracture types [4]. Repairing these bone defects caused by fractures, tumors, or bone cancer is still difficult in orthopedic surgery. In the current scenario, there is a growing demand for bone implants, with over 400,000 bone-grafting performed yearly in Europe and more than 600,000 in the United States. As reported in the orthopedic Industry Annual Report and Global Data's ORTHOWORLD report, the global sales of orthopedic products crossed \$55.5 billion in 2022, increasing from \$47 billion in 2020 amid two years of disruption stemming from the COVID pandemic [5]. This number is expected to increase dramatically over the coming decades as life expectancies rise [6].

Orthopedic implants play a crucial role in fracture treatment, which helps restore bone alignment and promote healing. In orthopedic implants, an artificial device called a porous scaffold is used to replace or reform the damaged bone (6–18 cm) and act as a new substitute therapy [7]. The geometric characteristics of a scaffold, such as porosity, pore size, and morphology, must be kept in mind during scaffold design. They directly affect the scaffold performance, including mechanical (compressive strength and Young's modulus) and biological properties (cell adhesion/proliferation, permeability, and surface area). The osteoblasts that form bone tissue prefer larger pores of 100–200 μm for bone regeneration after implantation [8]. When pores are very small, cell migration is constrained, leading to the development of a cellular capsule along the surface of the scaffold. Subsequent studies have revealed that pores size exceeding approximately 300 μm are essential for bone ingrowth and vascularisation. Conversely, pores with a size smaller than ∼300 μm were essential for encouraging osteochondral ossification [9]. Nevertheless, the proper pore size range remains uncertain. Additionally, scaffolds with uniform pore size may demonstrate effective performance compared to scaffolds with heterogeneous pores, which is yet to be clarified. The factor that affects the mechanical properties and surface area of a scaffold is the porosity percentage. The mechanical properties (Young's modulus and compressive strength) decline with increasing porosity percentage [8]. Qingyang *et al* fabricated PLA-based scaffold with varying porosity percentages (15%–45%) and pore shapes (regular, diamond, and gyroid) [10]. The study reveals that when porosity increases from 15%–45%, the compressive strength and Young's modulus decrease from 22 MPa to 7 MPa and 240 MPa to 110 MPa for a regular pore scaffold structure. Similarly, for the diamond pore scaffold structure, the compressive strength and Young's modulus decreased from 38 MPa to 17 MPa and 360 MPa to 220 MPa for the same percentage of porosity increment. Also, the gyroid pore scaffold observed the same trend, and compressive strength, and Young's modulus decreased from 32 MPa to 17 MPa (closer to the diamond pore scaffold) and 305 MPa to 280 MPa, respectively. The porosity is directly connected with surface area, and as the porosity increases, the surface area to volume ratio (SA/V) increases gradually. Matthew *et al* fabricated a Fe – Mn-based scaffold using diamond, gyroid, and primitive TPMS minimal surface. The result indicated that when porosity increases from 42 to 72%, the SA/V of gyroid, diamond, and primitive scaffold also increase from 12–22 $mm^2\ mm^{-3}$, 15–21 $mm^2\ mm^{-3}$, and 14–21 $mm^2\ mm^{-3}$, respectively [11]. Thus, it may facilitate the transportation of nutrients and oxygen, which may grow more cells inward [12]. Another factor is the geometry of the scaffold, which plays an essential role in scaffold performance. The scaffold with simple architecture is susceptible to collapse under high stress. In contrast to a scaffold with complex architecture, it enables them to recover after deformation and retain its elastic state [8].

Scaffold designs can be classified based on unit-cell designs. The unit cell refers to a pattern block repeated throughout the scaffold. The unit cell designs are characterized by parametric mathematical equations that define unit cell geometry with specific pore shape and size [13]. Recently, based on a unit cell design named triply periodic minimal surface (TPMS), it has attracted attention to design porous scaffolds for bone regeneration. TPMS minimal surface have quite high SA/V (compared to regular lattice), and its non-self-intersecting surfaces improve scaffold mechanical response and reduced stress concentration, depending on the shape and size of a scaffold [14]. Sanjairaj *et al* fabricated TPMS based primitive porous scaffold with unit cell size range of 1.5–3.5 mm and thickness range of 0.2–0.7 mm [15]. The result indicates that, for a unit cell size of 2.5 mm at 0.4 mm wall thickness, it exhibited maximum Young's modulus and fatigue strength of 4GPa and 100 MPa, respectively. As the unit cell size and wall thickness changed to 3.5 mm and 0.7 mm, the Young's modulus and fatigue strength decreased to 2.5 GPa and 40 MPa. Chenxi Peng *et al* conducted a study based on gyroid lattices with varying unit cells, from a single unit cell to a $6 \times 6 \times 6$ lattice. The study reveals that the stress-strain curve remains consistent regardless of the unit cell number. However, Young's modulus and plateau stress increased significantly when moving from a single-unit cell to a $2 \times 2 \times 2$ lattice. These properties then decreased when shifted to a $3 \times 3 \times 3$ lattice and stabilized with further increases in unit cell number [16]. Similarly, Almesmari *et al* studied the BCC plate lattice and primitive lattice designed in $2 \times 2 \times 2$, $3 \times 3 \times 3$, and $4 \times 4 \times 4$ arrays with a unit cell size of 14 mm and a relative density of 0.40. The results indicate a very similar elastic response and post-yielding behavior across the different lattice configurations and revealed that the lattice configuration of $2 \times 2 \times 2$ is adequate for analysis [17]. Recently, Jitendra *et al* fabricated IWP and diamond TPMS lattice with varying porosities of 50 to 80% at a unit cell size of 2 mm and evaluated mechanical properties and SA/V ratio [18]. The result shows that compressive strength decreases (258 to 84 MPa for IWP and 215 to 72 MPa for diamond) and the SA/V ratio increases (18 to 42 $mm^2\ mm^{-3}$ for IWP and 20 to 48 $mm^2\ mm^{-3}$ for diamond) with porosity (50 to 80%). Thus, to construct bone-mimicking scaffolds, control over unit cell size, porosity, pore size, and surface area is a major concern of researchers [19]. Various TPMS minimal surfaces exhibited distinct pore characteristics and surface areas at the same unit cell size and porosity. Also, the specific geometry of TPMS influences load distribution and stress generation within the scaffold. These scaffolds can be fabricated from different materials and fabrication techniques, depending on requirements such as compressive strength, porosity %, and SA/V, etc. The present study uses acrylonitrile butadiene styrene (ABS) resin as a scaffold material and fabricated 3D scaffold through stereolithography (SLA) technique. The study focuses on designing scaffold structures to determine the optimal unit cell size, required porosity percentage, and SA/V ratio (which results from different architectures). Also, it investigates the variation in deformation within the unit cell size and how deformation changes when these unit cells are stacked together. Thus, systematically characterize the TPMS minimal surface and identify the best suitable for a compression test experiment.

2. Materials and method

TPMS lattices are mathematical models of complex three-dimensional structures with minimal surface area when repeated in three directions. TPMS structures can mimic the complex interconnected porous structure of natural bone. These structures are repeated periodically in three dimensions while minimizing surface area within the boundaries. The word 'minimal surface' refers to surfaces where the mean curvature equals zero at every point. The pore size and porosity of TPMS scaffolds can be controlled by adjusting the mathematical parameters (unit cell size and porosity), allowing for customized 3D scaffold design [20]. The well-known TPMS surface, mainly gyroid, diamond, IWP (I-graph-wrapped package), primitive, Neovius, Fischer-Koch, F-RD (face-centered cubic rhombic dodecahedron), and PMY, are designed for bone tissue engineering applications. The implicit iso surface equations for each TPMS minimal surface are given as follows:

TPMS surface	Iso-surface equation	Equation no.
Schoen-Gyroid	$\sin(P_x x)\cos(P_y y) + \sin(P_y y)\cos(P_z z) + \sin(P_z z)\cos(P_x x) = t$	(1)
Schwarz-Diamond	$\cos(P_x x)\cos(P_y y)\cos(P_z z) - \sin(P_x x)\sin(P_y y)\sin(P_z z) = t$	(2)
Schoen−IWP	$2(\cos(P_x x)\cos(P_y y) + \cos(P_y y)\cos(P_z z) + \cos(P_z z)\cos(P_x x)) - (\cos 2(P_x x) + \cos 2(P_y y) + \cos 2(P_z z)) = t$	(3)
Neovius	$3(\cos(P_x x) + \cos(P_y y) + \cos(P_z z)) + 4(\cos(P_x x)\cos(P_y y)\cos(P_z z)) = t$	(4)
Schwarz Primitive	$\cos(P_x x) + \cos(P_y y) + \cos(P_z z) = t$	(5)
Fischer Koch S	$\cos 2(P_y y)\sin(P_y y)\cos(P_z z) + \cos(P_x x)\cos 2(P_y y)\sin(P_z z) + \sin(P_x x)\cos(P_y y)\cos 2(P_z z) = t$	(6)
Schoen-FRD	$4(\cos(P_x x)\cos(P_y y)\cos(P_z z)) - (\cos 2(P_x x)\cos 2(P_y y) + \cos 2(P_y y)\cos 2(P_z z) + \cos 2(P_z z)\cos 2(P_x x)) = t$	(7)
PMY	$2\cos(P_x x)\cos(P_y y)\cos(P_z z) + \sin 2(P_x x)\sin(P_y y) + \sin(P_x x)\sin 2(P_z z) + \sin 2(P_y y)\sin(P_z z) = t$	(8)

here x, y, and z are spatial coordinates in three-dimensional cartesian coordinates. 't' is the level set constant that controls volume fraction and adjusts the shape of the TPMS geometries, 'P' represents periodicities such as $P_i = 2\pi\frac{n_i}{L_i}$, $i = x, y, z$, and 'n_i' rrepresent number of unit cell in x,y and z direction, 'L_i' is the length of the lattice (structure dimension) in the corresponding direction, which generally represents the size of the unit cell. For example, the TPMS unit cell and corresponding lattice geometry formed by stacking unit cells are shown in figure 1. In addition, structures can be formed by either thickening the minimal surface to generate sheet-based cellular structures or solidifying the volumes enclosed by the minimal surfaces to create skeletal-based cellular structures, depending upon the iso value constant 't' [21]. To create a lattice structure, the volume bounded by the minimal surface can be solidified either $f(x, y, z) > t$ or $f(x, y, z) < t$ to form a solid phase lattice. On the other hand, to create the sheet phase lattice, offsetting the minimal surface in two directions and solidifying the volume enclosed by $-t \leqslant f(x, y, z) \leqslant t$ [22, 23]. The sheet-based structures were reported to exhibit superior mechanical properties than those of the solid based (skeletal) lattice [20]. Therefore, the present study mainly focused on sheet-based TPMS lattice structures.

For each TPMS minimal surface, the unit cell size (1–7 mm) and porosity (60%–90%) are considered to study their effect on mechanical properties and SA/V ratio. The finite element analysis based on structural mechanics physics is built in COMSOL Multiphysics. The mesh element size and mesh type details are provided in appendix. The work focuses on periodic scaffold designs consisting of unit cells organized in all three axes to produce a desired lattice structure. The scaffold unit cell model was selected as the lowest characteristic volume element to help finalize the unit cell size for a whole lattice structure design. The predominant force operating on the human femur bone was compressive in nature. Therefore, the current study is solely on unidirectional compressive force. American Society for Testing and Materials (ASTM) D695

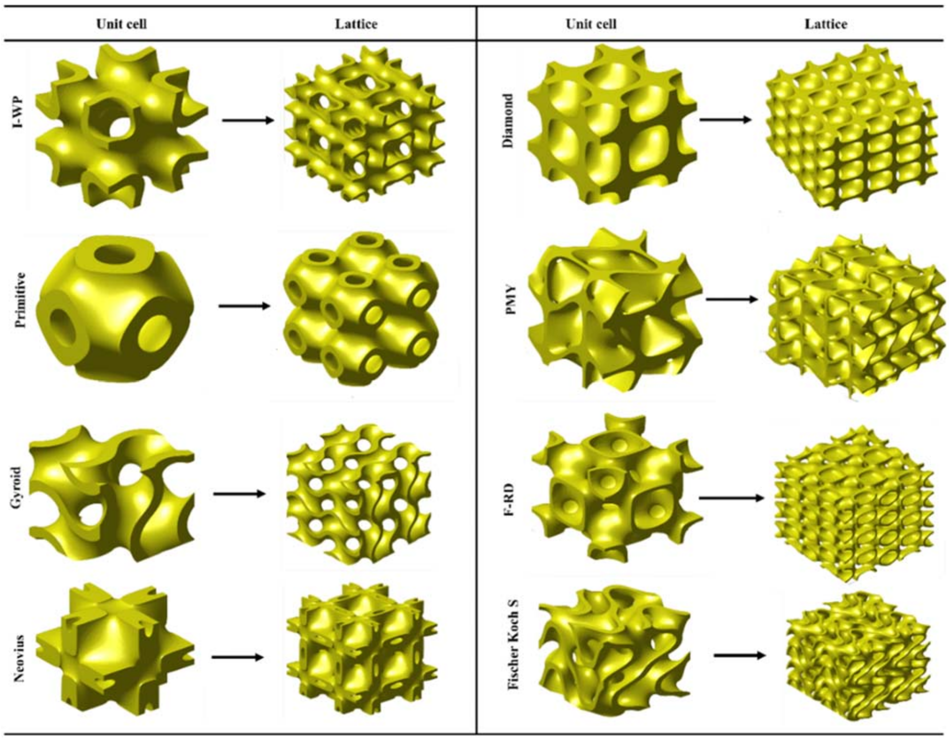

Figure 1. Types of TPMS unit cell and corresponding lattice structure with 70% porosity.

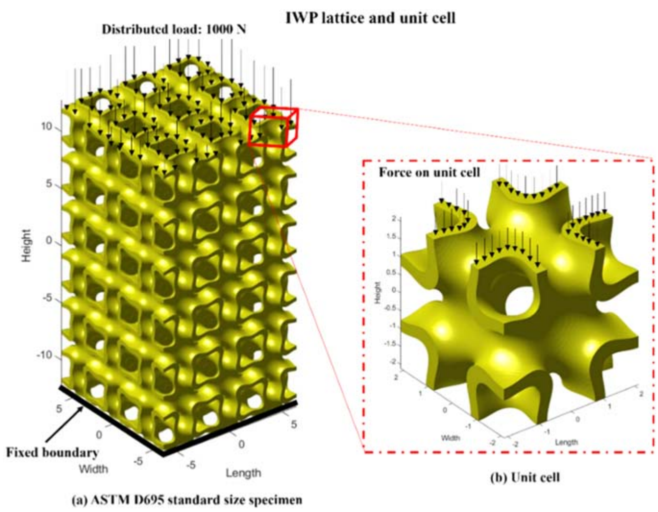

Figure 2. Schematic of loading and boundary condition of IWP (a) lattice and (b) unit cell.

standard size for compression test on polymers ($12.7 \times 12.7 \times 25.4$ mm^3) is used for compression simulation. A compressive force of 1000 N is applied to the lattice structure [24]. In the unit cell and lattice analysis, the force applied on individual unit cells is further divided based on the total number of unit cells on the surface of the lattice structure as listed in appendix, table A2. The scaffold material, namely medical-grade ABS resin, is considered for the simulation. It is known for its biocompatibility and strength, with the ability to return to its original shape after experiencing deformation. The material properties of medical-grade ABS resin are listed in appendix, table A3. The boundary condition for compression was referred from Oraib *et al* [25]. The load was applied on the top surface in a uniaxial downward direction, and the bottom surface was kept fixed for all the TPMS structures. For example, the boundary condition of the IWP unit cell and lattice geometry is shown in figure 2.

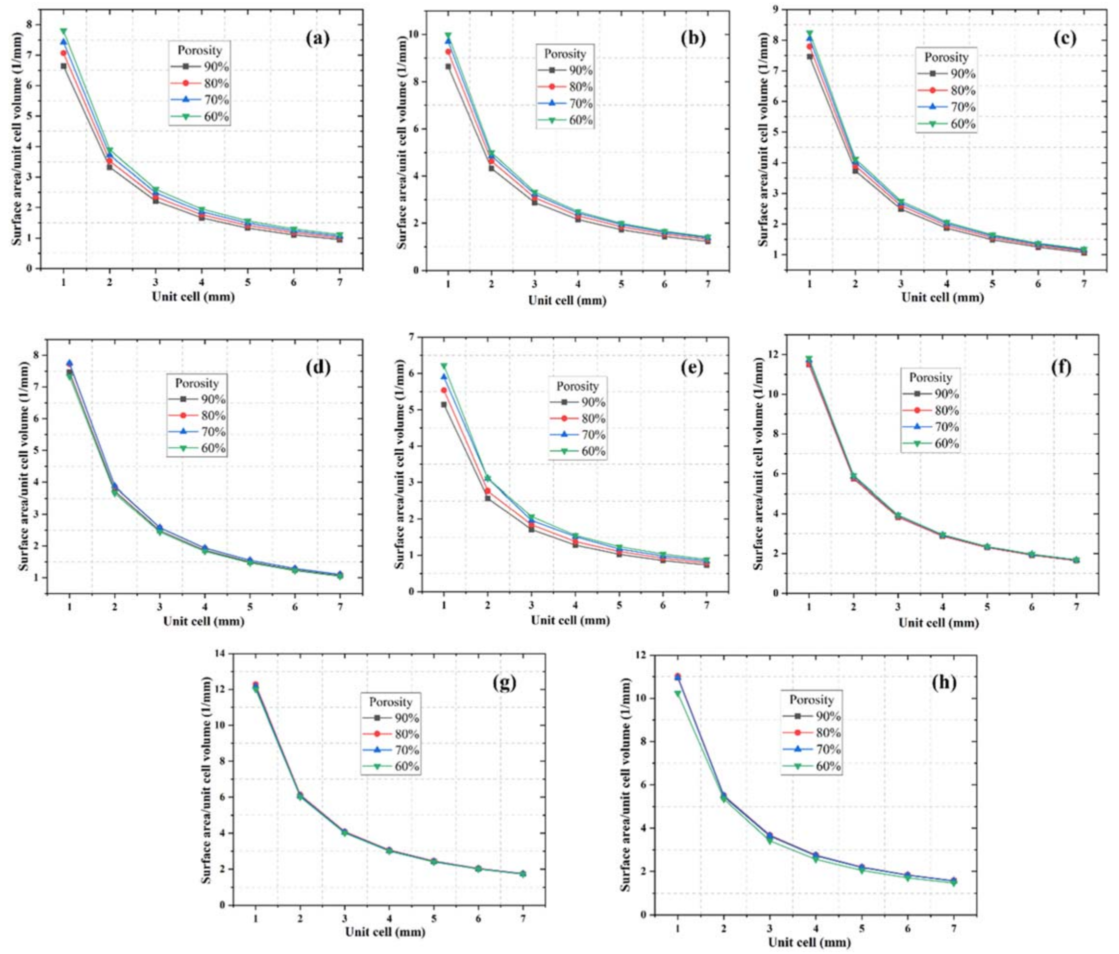

Figure 3. Surface area to unit cell volume ratio variations with unit cell size at different porosity percentages of TPMS unit cell (a) gyroid, (b) diamond, (c) IWP, (d) Neovius, (e) primitive, (f) Fischer Koch S, (g) F-RD, and (h) PMY.

3. Result and discussion

3.1. Unit cell size analysis

The surface area to volume ratio (SA/V) for each TPMS unit cell with varying unit cell size and porosity are represented in figure 3. The results indicate that the SA/V ratio of TPMS unit cells decreases with unit cell size increases 1 to 7 mm and vice versa. Similarly, when porosity increases by 60 to 90%, there is a slight variation in the SA/V ratio observed, and it is a negligible increment compared to unit cell size. A consistent trend emerges in SA/V among the TPMS structures for each unit cell size and porosity. Notably, F-RD and Fischer-Koch S structures exhibit maximum SA/V ratios (12 mm^2 mm^{-3}), and primitive structure shows a minimum SA/V ratio (6 mm^2 mm^{-3}) compared to other structures. This observation is primarily driven by the distinct geometrical attributes. When the unit cell size is small (1 mm), the TPMS exhibits intricate geometric characteristics, including smaller pore size (40–200 μm), resulting in a high surface area relative to its volume, which is beneficial for bone tissue engineering requiring large surface interaction. However, going for a smaller unit cell size may cause fluidity problems such as pores clogging, which can hinder fluid or cell movements through the lattice [9]. As the unit cell size increases from 1 mm to 6 mm, the surface area still increases but does not grow as much as the volume of the unit cell, which leads to a rapidly decreased SA/V ratio. However, beyond the unit cell size of 6 mm, the decrease in the SA/V ratio begins to slow down and stabilize, indicating that the SA/V ratio no longer meets the key requirement of a scaffold where the SA/V ratio is crucial for cell adhesion. To compensate for both fluidity and SA/V ratio, the optimized range of unit cell size as a parameter would be between 2 and 5. Additionally, this range of unit cell size makes it feasible to fabricate lattice structures without compromised mechanical properties, including compressive strength and Young's modulus.

The surface area of the unit cell is inversely proportional to the size of the lattice structure. This means that when smaller size unit cells are used to fill a fixed space to form a lattice, the resulting surface area within that volume will be higher. For example, figure 4 shows a cubical domain with a size length of 3 mm. If a 3 mm primitive unit cell is employed to fill the space, then one single unit cell will be required, and the resulting area

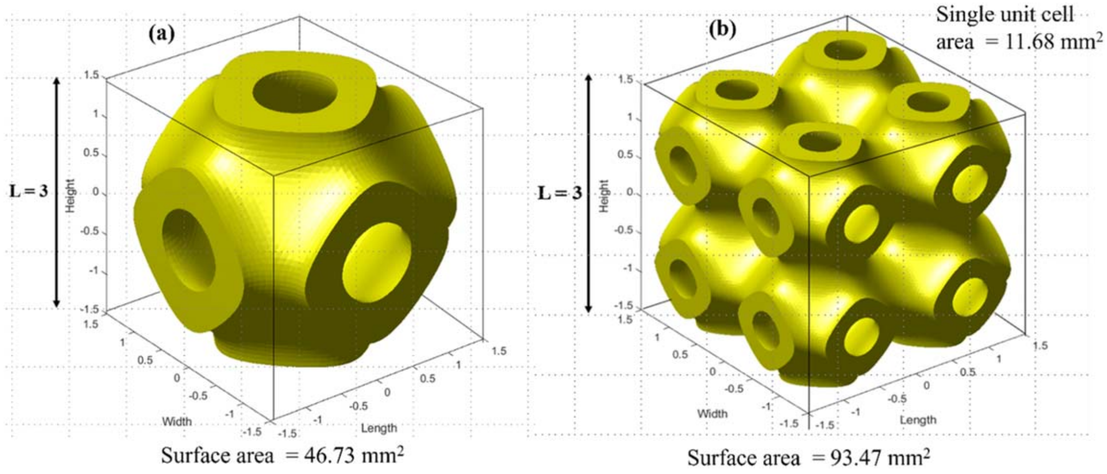

Figure 4. Surface area provided by a primitive TPMS minimal surface in a fixed volume, when unit cell size of (a) 3 mm and (b) 1.5 mm.

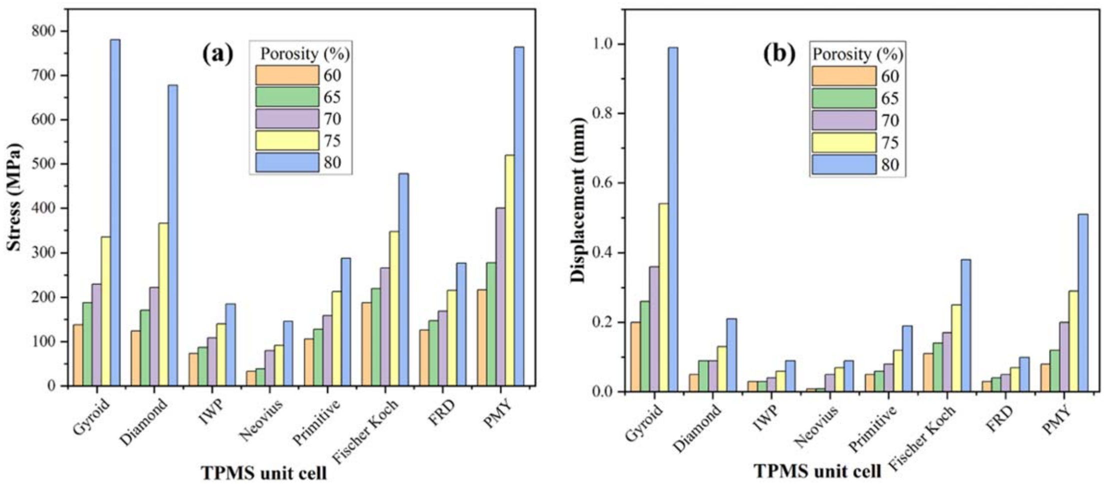

Figure 5. Variation of (a) Von Mises stresses and (b) displacement in TPMS unit cells of size 2.11 mm with different porosity percentages.

will be 46.73 mm^2 (figure 4(a)). Conversely, if the unit cell size reduced by half to 1.5 mm, then the total 8 unit cell will be required to fill the same space, and the corresponding resulting surface area will be doubled to 93.47 mm^2 (figure 4(b)). Therefore, in a fixed space, a smaller unit cell size requires a large number of the unit cell to fill it, resulting in a higher surface area [21].

3.2. Structural analysis

Structural analysis of structure can help understand how the structures behave under compressive force. One key aspect of structural analysis is evaluating the Von Mises stress distribution within the scaffold to predict failure. Additionally, the deformation in the structure can provide insight into how the scaffold responds to stresses and optimize the design characteristic for improved performance. As per earlier observation, the unit cell size should be within 2 to 5 mm. To obtain the ASTM D695 standard lattice, the unit cell sizes of 2.11, 2.54, 3.17, and 4.23 mm were considered to have a complete unit cell at the outer periphery of the lattice structure. Accordingly, the compressive force was applied on all the combinations of unit cell size as listed in appendix, table A2. Figure 5 depicts the von Mises stresses and displacements generated in unit cells of TPMS minimal surfaces of size 2.11 mm with different porosity percentages. For clarity and due to quite similar behavior, the von Mises stresses and displacement variation for the 2.54 mm, 3.17 mm, and 4.23 mm unit cell sizes are represented in figures A4(a)–(f), appendix. The stresses were varied, resulting from unique geometries, which provided distinct paths for stress distribution with the geometries. Also, as the unit cell size and porosity change, the geometrical characteristics of TPMS minimal surfaces change, which influences the stress pattern. Therefore,

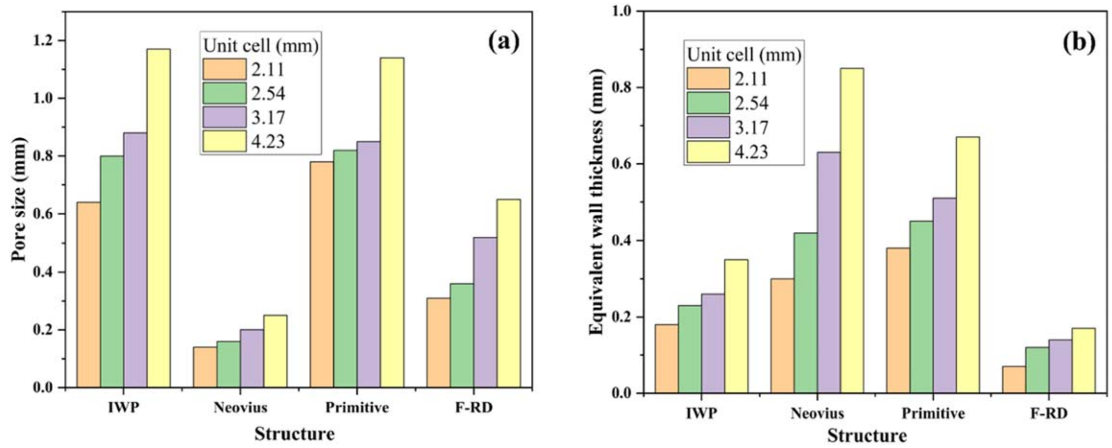

Figure 6. Variation of (a) pore size and (b) equivalent wall thickness of selected TPMS structure with unit cell size at 70% porosity.

at higher porosity levels, it potentially causes increases in the localized stress concentration due to a reduction in surface thickness to absorb and distribute stresses. At 60 to 65% porosity, the stress generated across the TPMS minimal surfaces is within the maximum strength of a femur bone. Meanwhile, the stress caused in IWP and Neovius at a higher porosity of 80% is also within the femur bone compressive strength. Thus, an appropriate porosity range could be between 70 and 80%. Moreover, the minimum stress generation was observed in Neovius, followed by IWP across the unit cell size and porosity. Conversely, it has been depicted that unit cell size does not show much effect on the stress generation, i.e., almost the same stresses were observed for all unit cell sizes at the same level of porosity (figure A4, appendix). The small changes in unit cell size (e.g., 2.11 mm to 4.23 mm) do not significantly alter the structure topology. Therefore, with small variations in unit cell size, the geometric pattern and topology (shape and interconnectivity) remain largely unchanged. This ensures that stress distribution within TPMS lattice structures remains relatively uniform across different unit cell sizes when the overall porosity percentage is constant. If talking from a general perspective, a structure with lower stress and deformation generation would be preferred for bone tissue engineering, which reduces the risk of mechanical failure. However, the smallest unit cell size of 2.11 mm has lesser stress generated in I-WP, Neovius, primitive, and F-RD TPMS minimal surfaces, where the resulting stresses are within the limit of femur bone strength at a porosity level of 70 to 80%.

Conversely, the displacement in the TPMS unit cells can vary based on several factors, including unit cell size, porosity percentage, force applied, and the material's properties in general. As the porosity percentage increases, the displacement may also increase. This is because a unit cell with higher porosity will have the thinnest surface and less resistance to deformation. However, the relationship between porosity and displacement is not always linear. It depends on the particular TPMS unit cell design characteristic as well. When a unit cell is subjected to compression force, the surfaces within the enclosed volume of a unit cell start to deform and move. This movement causes the unit cell to displace or change shape. At the microscopic level, the deformation of the surfaces occurs due to a combination of mechanisms, including elastic deformation, plastic deformation, and failure. Figure 5(b) represents the variation of displacement in the TPMS unit cell of size 2.1 mm with different porosity percentages. The displacement varied across the TPMS unit cells, and a maximum displacement of 1.97 mm was reported for the gyroid unit cell of size 4.23 mm at 80% porosity. As the unit cell size decreases to 2.11 mm, the maximum reported displacement for the gyroid is reduced to 1 mm at the same porosity level. On comparison, the analysis indicated that the minimum displacement of 0.01 mm was reported in Neovius, followed by IWP, primitive, and F-RD unit cell size of 2.11 mm and porosity of 60%–70%.

According to the analysis, the four distinct TPMS unit cells, namely I-WP, Neovius, primitive, and F-RD, exhibited stress and displacement under consideration at 70% porosity. These structures will be examined further to determine pore size and wall thickness variation with the unit cell size, as shown in figure 6. There is a proportional relation between pore size and wall thickness with the unit cell size. As the unit cell size increases by 2.11 to 4.23 mm, the pore size increases for each respective TPMS unit cell, which means that larger unit cell size exhibits larger pore size, which has a significant effect on surface area and compressive strength. Additionally, the wall thickness increases with the unit cell size, but the magnitude of this increase is less significant than observed for pore size. Moreover, it can be observed that at the same unit cell size and porosity, the TPMS unit cell exhibited distinct pore size and wall thickness, which is attributed to its unique surface architecture arrangement. Specifically, IWP and primitive consistently display large pore sizes (0.64–1.17 mm) across the entire range of unit cell sizes. On the other hand, Neovius and primitive unit cells reported larger wall thicknesses

(0.3–0.85 mm) for all unit cell sizes. This observation is essential in determining the structural characteristics of a scaffold. Additionally, this study conflicts with the assumption that increasing the unit cell size would lead to increases in the pore size, and a decrease in the wall thickness, as usual, was observed for regular lattice structures.

3.3. Lattice analysis

Since bone grafts that are made up of a combination of single-unit cells combined to form a lattice cell structure are known as scaffold-based grafts. Once the individual unit cells are created, they are combined to form a lattice cell structure. In appendix, figure A5 represents a schematic of a model lattice structure placed at the defects placed. This structure provides a large surface area for cell attachment and allows for the diffusion of nutrients and waste products throughout the graft. However, it is important to facilitate a comparative analysis between the lattice and individual unit cells to reveal the stress distribution and deformation within the geometry. Moreover, it helps in design improvement to meet the needs.

To avoid computational loads, the sample size is considered smaller than the ASTM standard size, and the compressive load is applied accordingly. Therefore, for the analysis, the sample size of $4.23 \times 4.23 \times 4.23$ mm^3 is considered with a lattice configuration of $2 \times 2 \times 2$ at a unit cell size of 2.11 mm and 70 % porosity, as shown in appendix, figure A6.

As per earlier discussion, the best TPMS unit cells, namely I-WP, Neovius, primitive, and F-RD, are considered for the analysis. Figure 7 represents the stress and displacement contour plot of a single unit cell and its corresponding lattice geometry. The stress generated in the IWP lattice is 2.75% less than the stress generated in its single-unit cell. This could be because lattices have a more stable geometric configuration than the unit cell. The interconnected complex network of surfaces in the lattice structure distributes force more uniformly, reducing the stress concentration points that occur in a single unit cell. Conversely, the displacement reported in a lattice structure is twice that in a single-unit cell. This is because the I-WP lattice structure comprises multiple unit cells that are connected to each other. The unit cell closer to the point of application of force experiences more deformation than the ones further away. This causes a redistribution of stress and strain throughout the lattice, increasing the overall displacement. Meanwhile, the deformation in a single unit cell is limited to the unit cell alone without influence from the neighboring unit cell.

In the case of Neovius, the stress generated in the lattice is 13.6% higher than the stress reported in a single unit cell. This is because the stress generated in a single unit cell mainly occurs in its middle. In contrast, the stress induced in the lattice does not influence greater due to the neighboring unit cell. It is not necessarily true that the stress generated in a Neovius lattice is always higher than the stress caused in its single-unit cell. However, there are specific scenarios where lattice reported higher stress than a single unit cell. For example, the Neovius lattice doesn't have much redundancy, where if any of the surfaces fails, the load will be redistributed to the remaining cell, potentially leading to higher stress concentration. Conversely, the lattice structure also reported a higher displacement of 0.07 mm, as shown in figure 7. It happened due to a compressive force distributed throughout the lattice, which causes the individual unit cell to undergo displacement initiated from the center of each connected unit cell. On the other hand, a single-unit cell will deform in a more localized manner with stress concentration in the middle area of the cell only.

In the primitive unit cell, the stress generated in the lattice structure is around 22.01% less than in a single unit cell. The interconnected surfaces in the lattice create a system of load paths that distribute stress more evenly and minimize the stress concentration. This will help to absorb the stress without concentrating on one specific area of the lattice, leading to geometric stability. Whereas the displacement in the lattice is higher than in a single-unit cell. Like the Neovius lattice, the primitive lattice comprises interconnected unit cells that create the complex internal architecture. When a force is applied, the load is distributed throughout the structure, and individual unit cells undergo deformation, as shown in figure 7. Therefore, it allows the lattice to withstand a larger amount of displacement without causing localized failure.

In the case of the F-RD unit cell, the stress generated in the lattice structure is 12.42% less than the stress induced in a single unit cell. It is because the F-RD lattice has centered cubic symmetry with multiple loading direction that allows for uniform distribution of a force throughout the lattice, as shown in figure 7. Also, the rhombic dodecahedral shape in the F-RD increases the surface area of the lattice, which allows greater load bearing capacity. This helps in reducing stress concentration in the lattice. At the same time, the reported displacement was observed to be higher than the displacement in a single-unit cell. Like other TPMS structures, the individual unit cell undergoes deformation to accommodate the force. Also, due to a complex structure, the lattice stiffness would be higher than a single unit cell, allowing interconnected surfaces to accommodate deformation.

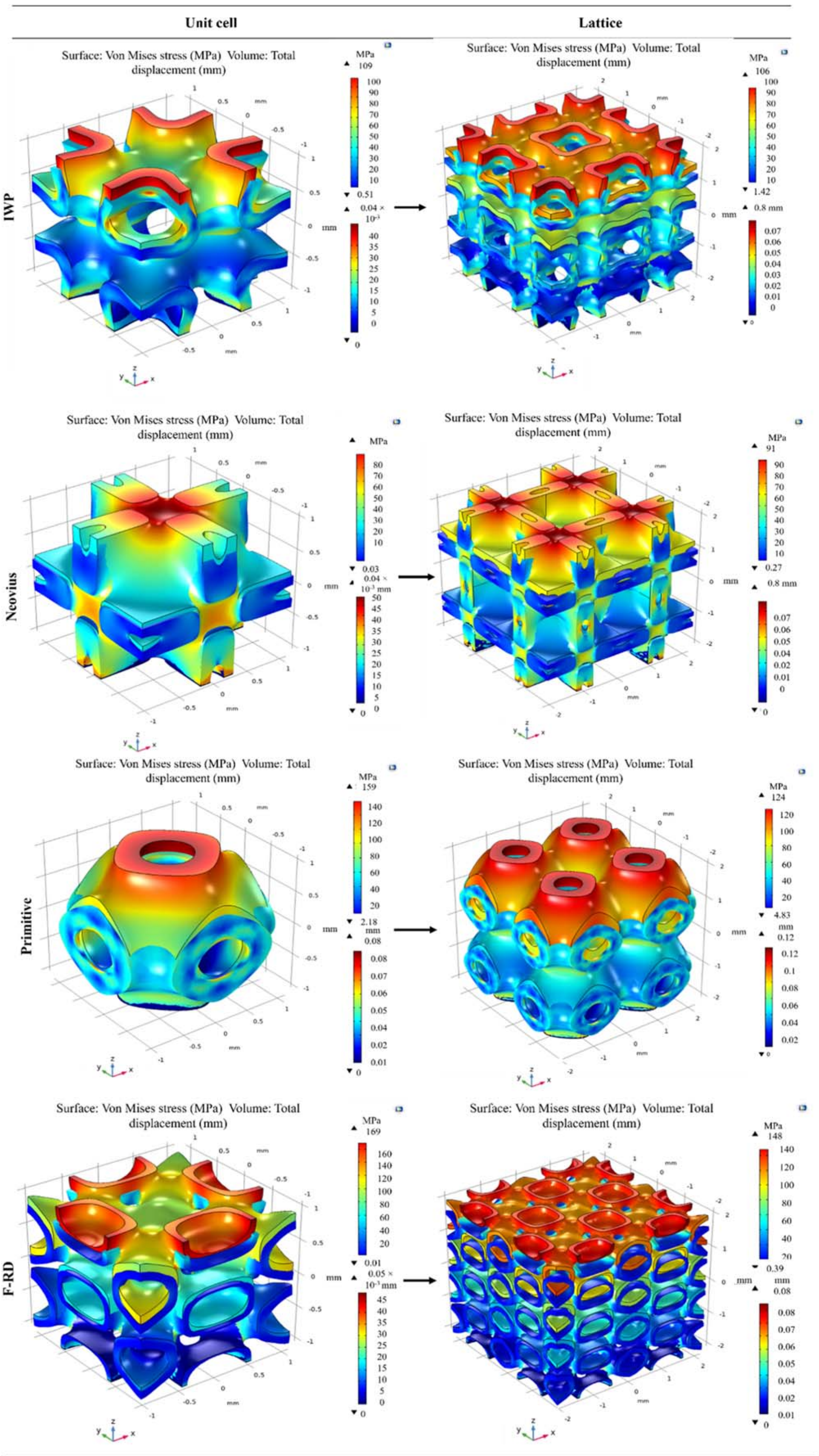

Figure 7. Stress and displacement contour plot of IWP, Neovius, primitive, and F-RD unit cells. The first column represents a single unit cell of size $2.11 \times 2.11 \times 2.11$ mm^3 and the second column represents a lattice structure of size $4.23 \times 4.23 \times 4.23$ mm^3.

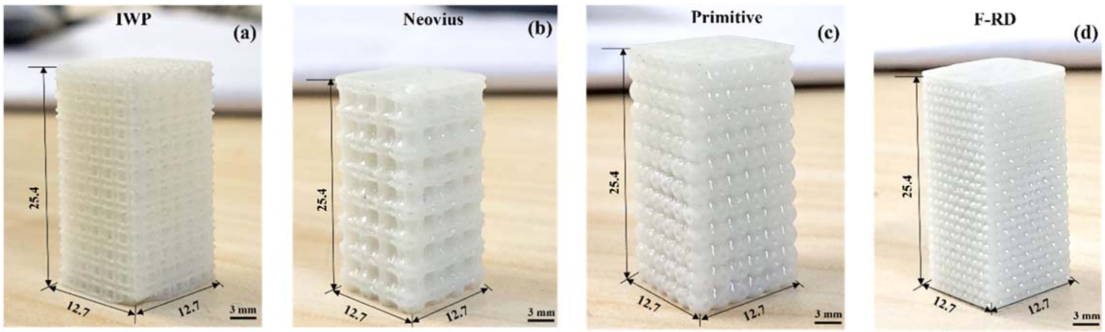

Figure 8. Fabricated scaffold of selected TPMS lattice structure as per ASTM D695 standard. (a) IWP, (b) Neovius, (c) primitive, and (d) F-RD scaffold. All dimensions are in mm.

Table 1. Geometric characteristics of selected TPMS lattice structure.

Lattice structure	Design pore size (μm)	Measured pore size (μm)	Design equivalent wall thickness (μm)	Measured equivalent wall thickness (μm)
IWP	640	624 ± 9	180	207 ± 2
Neovius	140	124 ± 5	290	300 ± 4
Primitive	780	758 ± 3	380	410 ± 3
F-RD	310	299 ± 7	70	110 ± 2

Overall, the lattice structure imparts higher stiffness and reduced stress concentration compared to a single-unit cell. Nevertheless, there is a need for the experimental study to understand further how TPMS lattice strength and deformation behavior occur under realistic conditions.

3.4. Fabrication and mechanical properties of printed scaffold

The TPMS structure I-WP, Neovius, primitive, and F-RD based scaffold were printed from medical grade ABS resin using SLA. SLA employs photochemical processes to cross-link chemical monomers and oligomers to generate 3D scaffolds. The scaffolds are designed and fabricated as per ASTM D695 standards, which specify the dimension of the sample as $12.7 \times 12.7 \times 25.4$ mm^3. The .stl files were sliced into a slicing software known as Flash print, wherein all printing parameters are adjusted for the fabrication as listed in appendix, table A3. Figure 8 shows the printed scaffold with a unit cell size of 2.11 mm and lattice size of $12.7 \times 12.7 \times 25.4$ mm^3 with 70% porosity. The structural behavior of the scaffolds is evaluated experimentally through the compression test. The experiment is conducted using the MTS-810 hydraulic compression system with a 100 kN load cell at 1.3 mm min^{-1} up to a strain level of 60%. The fabrication begins with selectively cured layers using UV light. This precise layer-by-layer construction creates the scaffold. However, during fabrication, the structure can have localized deformation as each layer solidifies. This deformation can be due to thermal effect material shrinkage, but it is usually minimal compared to other 3D printing techniques. Additionally, in fabricated scaffolds, there is a consideration for the potential pore clogging. The curing process can cause the resin to flow into the pores and solidify, which may need to be addressed during the design (unit cell size selection) or through post-processing steps such as chemical washing.

The pore size and equivalent wall thickness were measured for each selected TPMS lattice structure using a digital optical microscope. The dimensions of the pore size and equivalent wall thickness are summarized in table 1. The measured dimension of the structures slightly differs from the design dimension, which might be due to the fact discussed in the above paragraph. Figure 9 shows the design and measured dimension of the lattice structure of a unit cell size of 2.11 mm. The measured pore size is smaller than what was originally obtained in the design, while the measured equivalent wall thickness is greater than the design. These slight variations in dimensions are obvious in 3D printing-based structures and are in accordance with findings from previous studies [26, 27].

The stress-strain curve and resulting Young's modulus obtained from the compression test are shown in figure 10. During the compression, the scaffold typically follows a distinct phase. The process begins with the elastic limit, where the structure temporarily changes shape in response to the applied load. As the stress continues to rise, the structure enters the plastic region and undergoes a densification phase, which leads to the

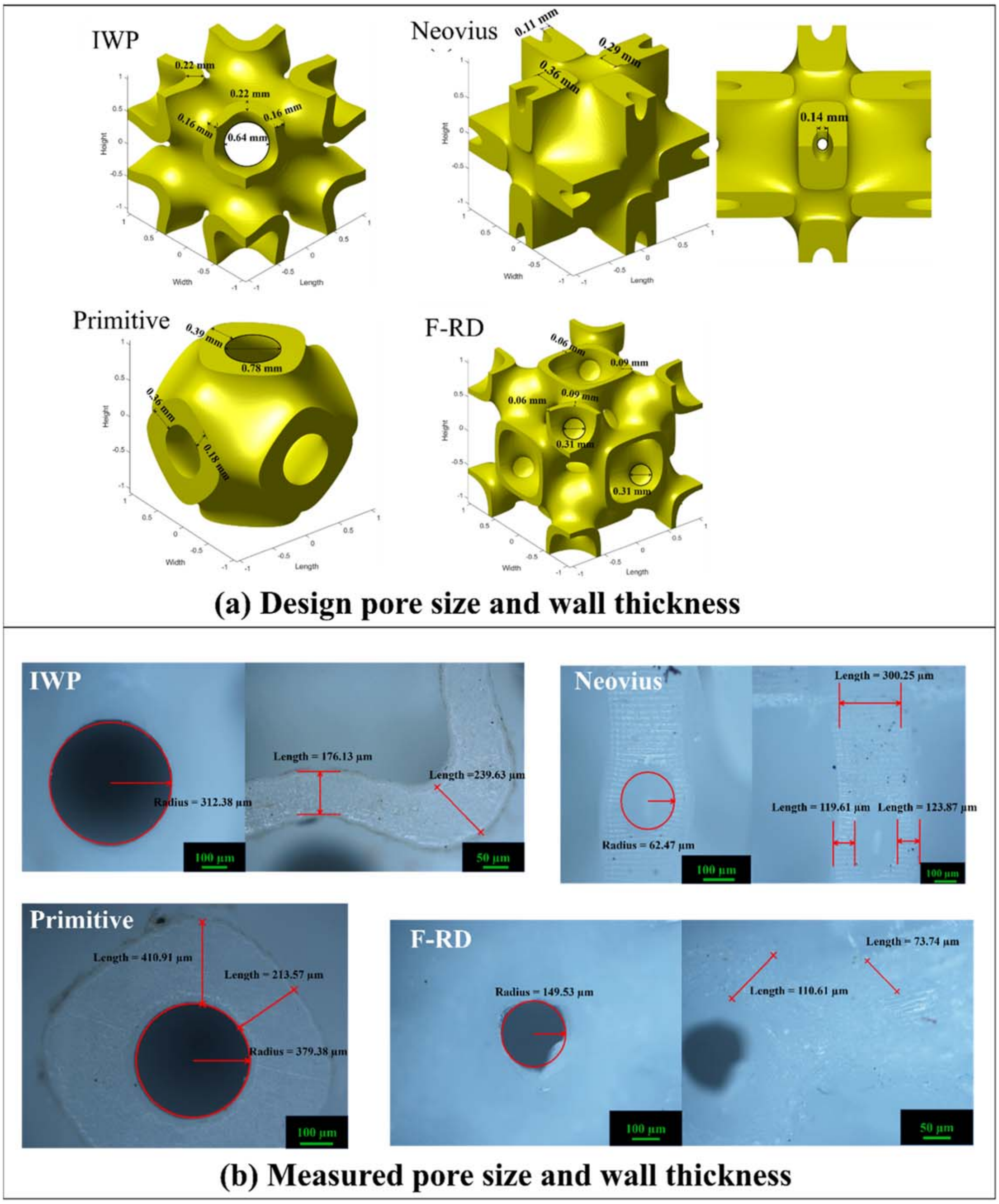

Figure 9. (a) CAD model representation and (b) optical images of the selected TPMS unit cells named IWP, Neovius, primitive, and F-RD lattice structures.

closing of the pore. Finally, when the structure's strength is suppressed, failure occurs, often marked by cracking and collapse for all the TPMS structures.

Comparing the stress-strain behavior of the selected TPMS lattice, it can be seen that for the IWP structure, the yield strength of 39.86 MPa was observed at 4.2% of yield strain, then the densification starts, and the structure visibly failed at nearly 35.46%. However, the highest stress is achieved among the structures prior to collapse. In contrast, the Neovious and F-RD structures show a similar trend of stress-strain curves. When the structure reached the first stress peak, resulting in a significant reduction in stress and the structure unable to resist the applied load, it failed at nearly 33.6% of the strain level. In the case of primitive, the lowest yield strength, 23.94 MPa, was reported at 3.8% of yield strain; then, after the densification, the structure reached the peak stress of 56.85 MPa, and the structure failed nearly at a strain level of 28.7%. Table 2 lists the obtained mechanical properties of the scaffold. It is evident from the observation that the compressive yield strength and Young's modulus of all TPMS structures are closely similar to that of native cancellous bone [28]. However,

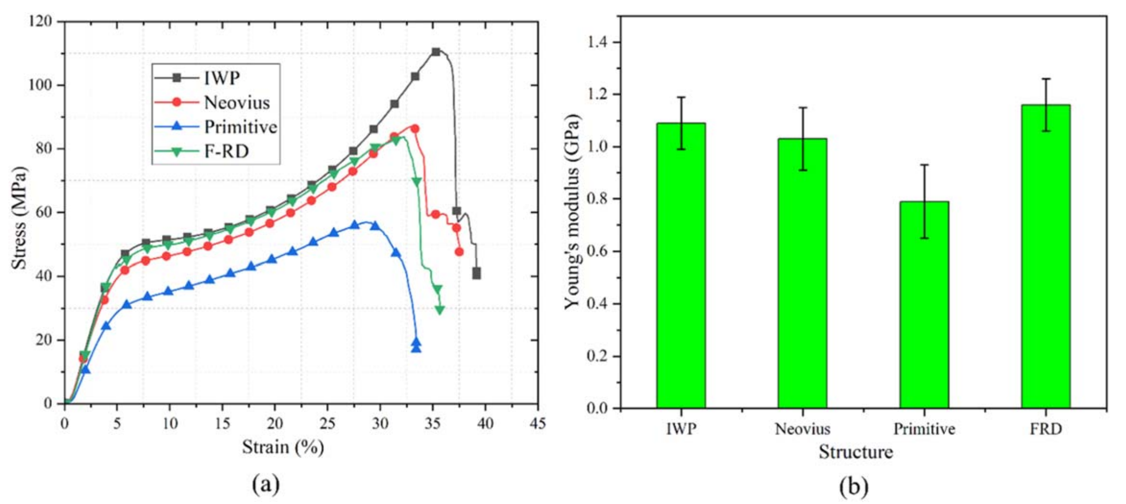

Figure 10. (a) Stress-strain plot and (b) reported Young's modulus of the structures during the compression test.

Table 2. Mechanical properties of selected TPMS structure under compression test.

Structure	Yield strength (MPa)	Yield strain (%)	Yield load (kN)	Peak stress (MPa)	Peak load (kN)
IWP	39	4.26	6.42	111 ± 2	18
Neovius	33	3.90	5.40	87 ± 5	14
Primitive	24	3.88	3.86	57 ± 2	9
F-RD	39	4.15	6.25	84 ± 1	13

when examining the pore size, it becomes apparent that the IWP and primitive structures have similarities with the cancellous bone pore size [29]. In contrast, the Neovious and F-RD structures exhibit a range of pore sizes nearly similar to cortical bone [29]. Notably, the yielding load of all the TPMS structures surpasses the load-bearing capacity of bone, capable of sustaining up to 1kN of load, which is considered below what may be physiological for numerous daily activities; however, it kept the scaffold within the elastic limit and prevent damage [24, 30].

The photograph of the deformation (highlighted by red line) behaviour of the TPMS structure at different strain levels is shown in figure 11. The layer deformation pattern was observed and proceeded through the layers beginning from the strut and wall surface. As the compression proceeds further, the deformation starts to move from one strut/wall surface to another. As the deformation reaches the side end of the test sample, it reflects and transmits the deformation to the adjacent strut/wall surface and starts to deform. However, this way, the deformed layers proceed and are repeated till the sample starts densification. For the IWP structure, the deformation starts from the top and progresses row by row, where the bulging deformation zone is formed at nearly 5%–15% of the strain level. On further deformation, the crack begins to appear at 25% of the strain level, resulting in convex pattern deformation. Finally, a brittle fracture was observed at nearly 35 % of the strain level. In the case of Neovius, the bulging deformation was initiated at 5% of strain level and propagate with curve deformation zone till the 15% of strain level. As the deformation increased, the crack along the strut became more frequent, and deformation changed to a curved deformation zone, where the lattice structure failed at nearly 33 % of the strain level. For primitive lattice, the deformation pattern starts to form row by row along with the bulging in distinct axes at 5%–15% strain level. At higher deformation, the bulging deformation zone changed to a curved and crack deformation zone where row by row deformation pattern vanished altogether at nearly 25%–28% of the strain level. In the F-RD lattice structure, the deformation starts with a crack formation at the top surface. This crack formation propagates with an apparent bulging zone in the structure up to a strain level of 15%. Furthermore, the crack-bulging zone changes to the crushing zone as the deformation is highly localized, nearly at 32% of the strain level, resulting in a rapid fall in stress value.

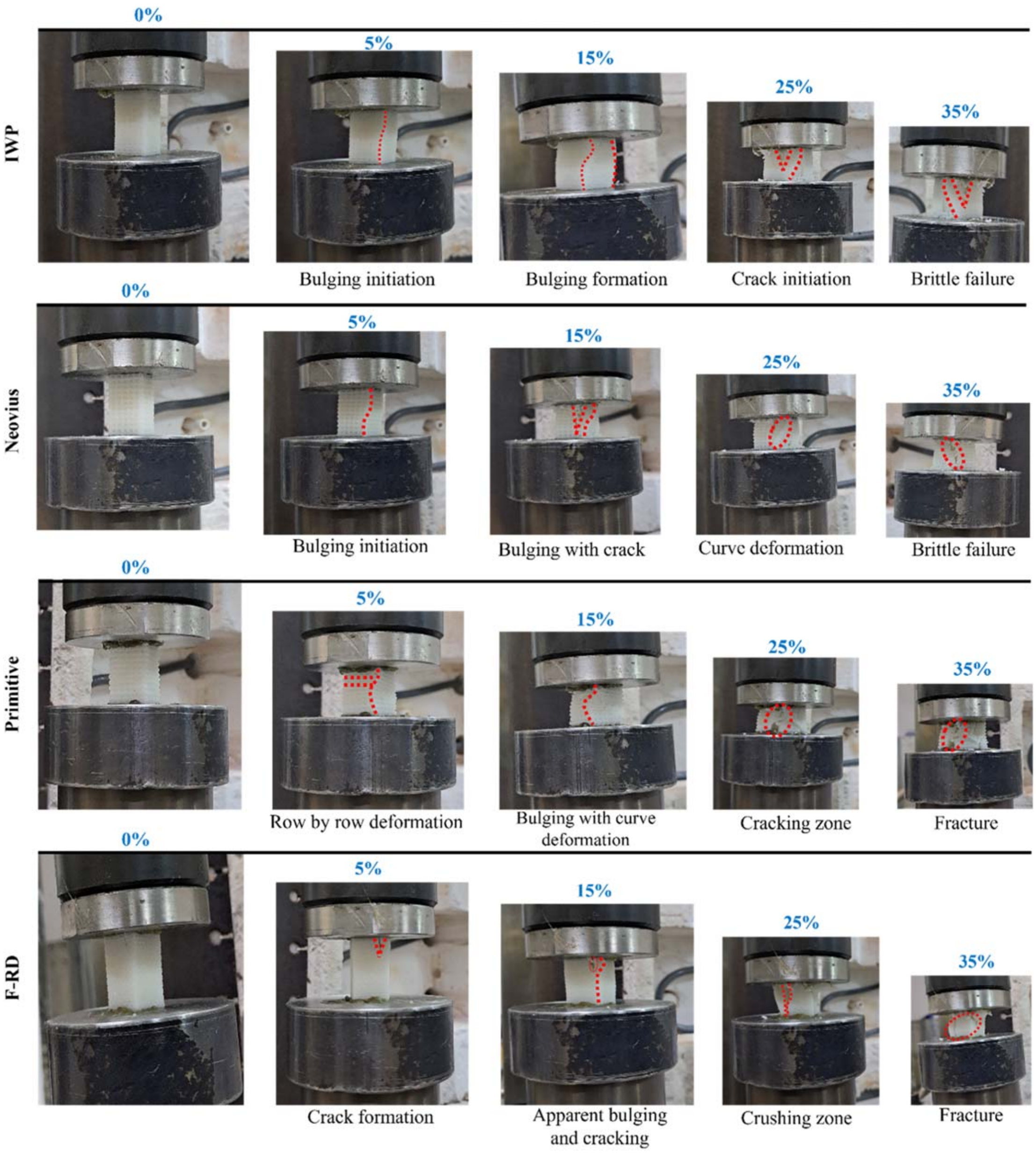

Figure 11. Deformation behaviours at different strain levels during compression test of IWP, Neovius, primitive, and F-RD lattice structure.

4. Conclusion and future scope

The present study provides the foundation for scaffold design on TPMS unit cell structure. A wide range of TPMS structures were subjected to computational analysis to determine the effect of the structure's characteristics, such as design, unit cell size, porosity percentage, and surface area to volume ratio on stress generation within the structure. Furthermore, the structures were compared based on unit cell size and structural analysis to determine the best four TPMS unit cells, namely, IWP, Neovius, primitive, and FRD, for further analysis. A comparative study was made between the lattice and individual unit cells to understand the stress distribution and deformation within the geometry. Also, experiments were conducted to analyze the mechanical properties and deformation behavior of the structures.

The significant findings and observations from this study are as follows:

1. SA/V was inversely proportional to the unit cell size, and the ratio decreased slowly beyond the 6 mm unit cell size. Thus, selecting a unit cell size greater than 6 mm would not be worthwhile.

2. The unique TPMS geometries play a significant role under a compressive force and have distinct paths for stress distribution. The maximum compressive strength and Young's modulus of 39.8 MPa and 1.09 GPa were reported in IWP, followed by Neovious and F-RD lattice structures.
3. Lattice structures have a more stable geometric configuration than a single-unit cell. The interconnected surface network helps to distribute force and reduce stress concentration. Thus, using multiple unit cells to form an enclosed network within a fixed volume is more efficient than employing a larger single-unit cell.
4. The small changes in unit cell size (2.11 mm to 4.23 mm) do not significantly alter the structure topology. This stability in structure design ensures that stress distribution within TPMS lattice structures remains relatively uniform across different unit cell sizes when the porosity percentage is constant.
5. The distinct geometric features of TPMS unit cells result in varied deformation behaviors under the same compression conditions. In IWP and primitive lattices, deformation begins at the top and progresses row by row, forming a bulging deformation zone. Conversely, Neovius and F-RD lattices exhibit deformation starting with a bulging and crack formation zone, which further transitions into a crushing zone as deformation progresses.

The current study analyzed uniform TPMS structures to understand their geometric characteristics. In the near future, a promising design approach should be utilized to design a functionally graded TPMS structure (FGS) that might enable us to create a gradient in mechanical properties to give a more accurate representation of a native bone. It can be achievable by varying the unit cell size or relative density along the spatial direction, in which pore size and porosity range need to be controlled. Similarly, hybrid TPMS structures can come up with a collective potential to deal with the anisotropy of a native bone. The maximum compressive strength exhibited by IWP and Neovius TPMS unit cells can be designed into a hybrid structure by merging them in specific patterns. The hybrid structures can be created with or without spatial gradient to control physical and geometrical characteristics. These hybrid structures have the potential to enable customization on biomechanical properties such as compressive strength, Young's modulus, SA/V ratio, permeability, and wall shear stress.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

CRediT authorship contribution statement

Raj Kumar: Conceptualization, Investigation, Formal analysis, Methodology, Data curation, Writing-Original draft preparation. **Janakarajan Ramkumar:** Conceptualization, Methodology, Resources, Supervision, Writing-Reviewing and editing. **Kantesh Balani:** Conceptualization, Methodology, Supervision, Writing-Reviewing and editing.

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Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

Appendix

As discussed earlier, the unit cell size should be 2–5 mm. Although to have the complete unit cell at the outer periphery of the ASTM standard lattice, the unit cell sizes of 2.11, 2.54, 3.17, and 4.23 were considered. The corresponding lattice configuration and force applied are listed in table A2. In COMSOL Multiphysics, mesh setting window, sequence types: physics controlled mesh, element size: fine, and mesh type: tetrahedral mesh is applied. The maximum mesh element size varies from 0.172–0344 mm, and the minimum element size varies from 0.0215–0.043 mm. The total number of elements used in the simulations ranged from 12265 to 67311.

Therefore, the specific geometry of each TPMS unit cell influences the structures' deformation behaviour. Each geometry possesses unique structural characteristics that decide the strut surface distribution and their connectivity within the lattice. Through the adjustment in structural characteristics, it becomes possible to optimize the compressive yield strength and more efficient distribution of stress and reducing the localized stress concentration within the lattice structures.

Table A1. Structural and mechanical properties of human bone tissue [2, 31].

Mechanical properties	Cortical bone	Cancellous bone
Compressive strength (MPa)	100–230	2–12
Tensile strength (MPa)	50–150	10–20
Strain to failure (%)	1–3	5–7
Fracture toughness (MPa.m$^{1/2}$)	2–12	
Young's modulus (GPa)	7–30	0.5–0.05
Porosity (%)	5–10	40–95
Surface/bone volume (mm^2/mm^3)	2.5	20
Density (g/cm^3)	1.99	0.05–0.60

Table A2. Lattice structure configuration.

Unit cell size (mm)	Sample size (mm^3) as per ASTM	Lattice configuration	Number of unit cells	Force on unit cell (N)
2.11	$12.7 \times 12.7 \times 25.4$	$6 \times 6 \times 12$	432	27.77
2.54	$12.7 \times 12.7 \times 25.4$	$5 \times 5 \times 10$	250	40.00
3.17	$12.7 \times 12.7 \times 25.4$	$4 \times 4 \times 8$	128	62.50
4.23	$12.7 \times 12.7 \times 25.4$	$3 \times 3 \times 6$	54	111.11

Table A3. Material properties of medical grade ABS resin and SLA printing parameters.

Properties	Value
Density (g cm^{-3})	1.2
Young's modulus (GPa)	2.08
Poisson's ratio	0.35
Printing parameters	
UV beam wavelength (nm)	405
Layer thickness (μm)	50
Exposure time (s)	10

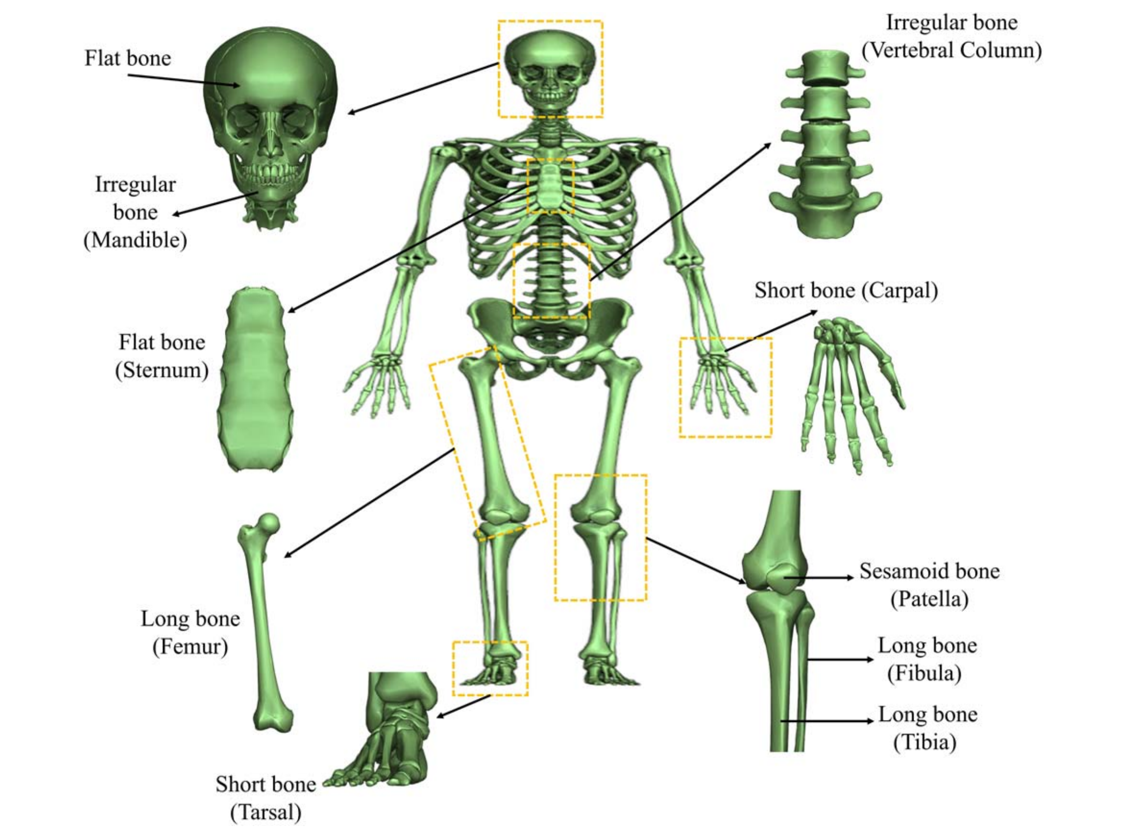

Figure A1. Anatomical location of different bones in the human body.

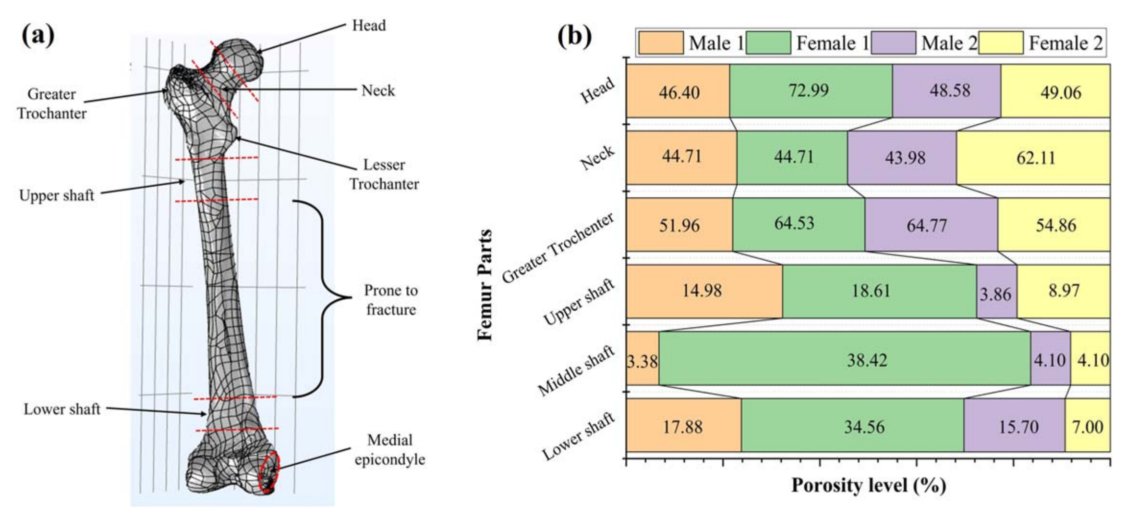

Figure A2. (a) femur bone's parts section, (b) represents the percentage of porosity available in different parts of a femur bone. Modified from Ref [3].

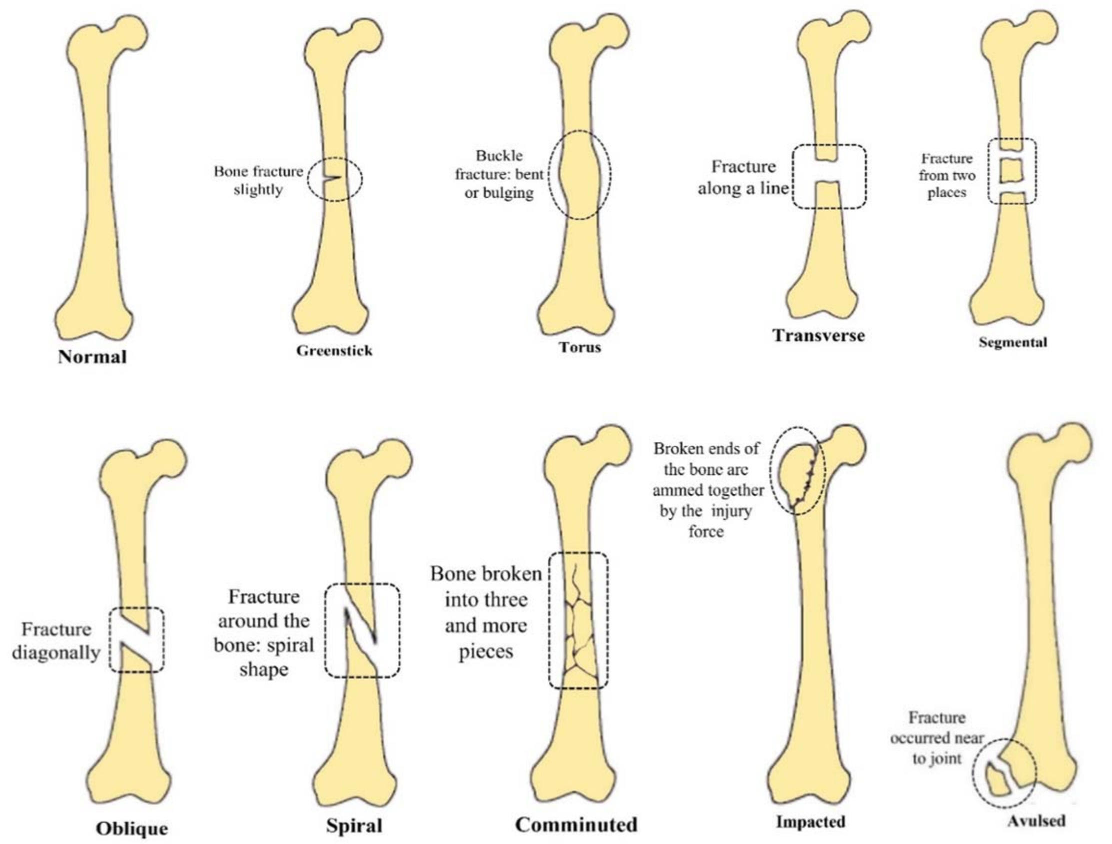

Figure A3. A common bone defect that prevails in orthopedics. Reprinted from MSD manual, consumer version: https://www.msdmanuals.com/en-in/home/injuries-and-poisoning/fractures/overview-of-fractures [4].

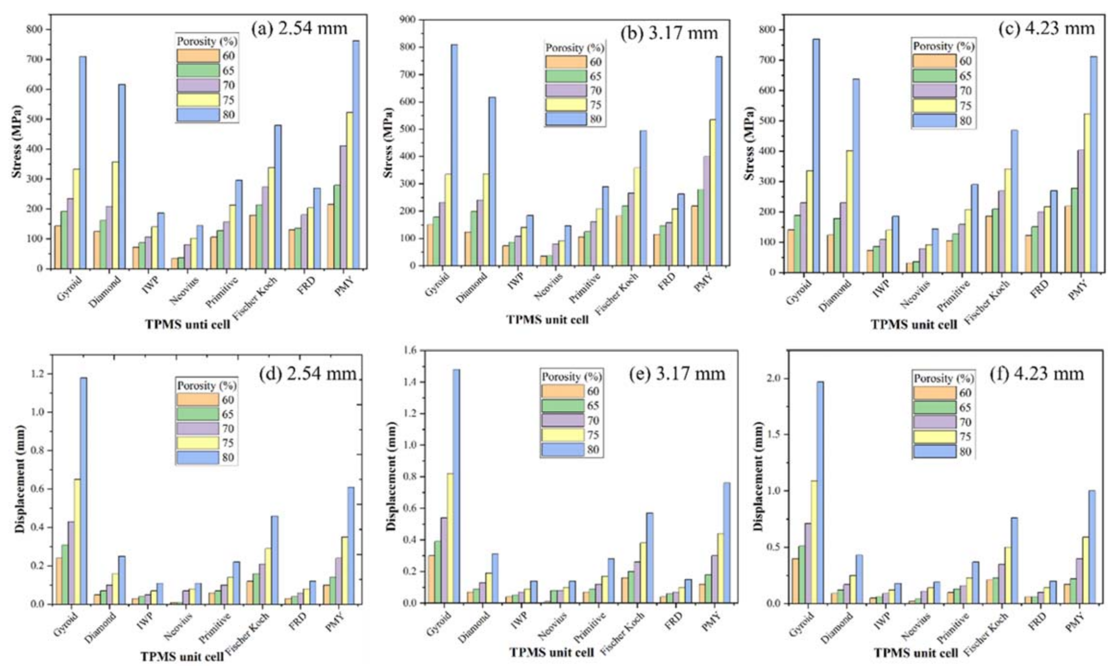

Figure A4. Variation of von Mises stresses (a)–(c) and displacement (d)–(f) with varying porosity percentages for a unit cell size of 2.54 mm, 3.17 mm, and 4.23 mm.

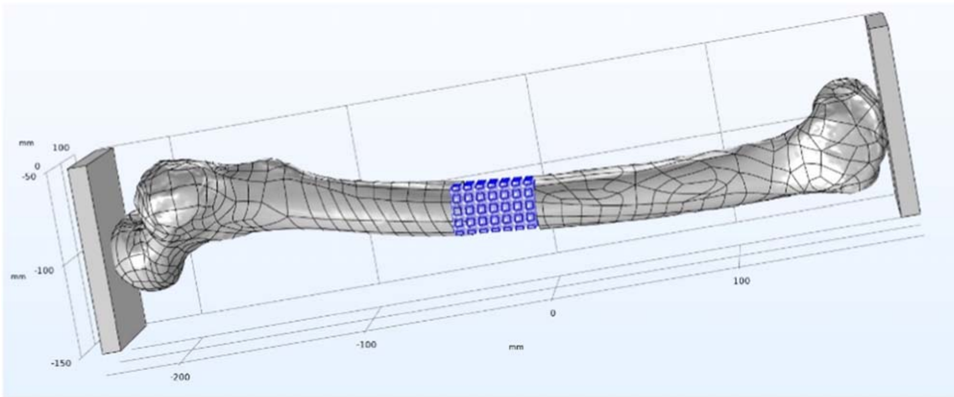

Figure A5. Implanted scaffold at segmental bone defects site. Courtesy: https://cults3d.com/en/orders/66400318.

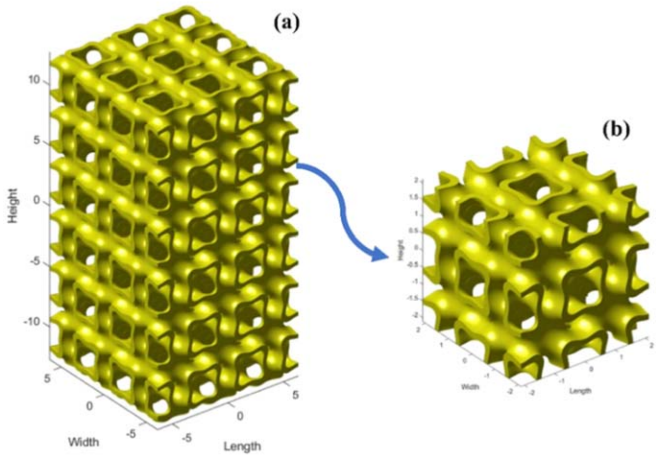

Figure A6. (a) ASTM standard specimen 12.7 × 12.7 × 25.4 mm^3, (b) sample size 4.23 × 4.23 × 4.23 mm^3 with lattice configuration 2 × 2 × 2 with unit cell size 2.11 and 70% porosity.

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